Date _____

Expt. No. 1 Page No. 1.

Aim:- To produce a single pixel of pre-specified pattern on screen.

Source Code:-

```
#include <graphics.h>
#include <conio.h>
main()
{
 int gd = DETECT, gm;
 initgraph(&gd, &gm, "C:\\TC\\BGI");
 putpixel(25,25, RED);

 getch();
 closegraph();
 return 0;
}
```

Viva Question:-

Q1 What is Pixel?

Ans A pixel (Picture Element) is the smallest unit of a digital image displayed on a screen. Each pixel represents a specific color & brightness value.

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Q2 What is Bitmap?

Ans A bitmap is a digital images made up of a rectangular grid of pixels.
Each pixel's color is stored as binary data.

Q3 What is Refresh Buffer / Frame Buffer?

Ans A frame buffer is a memory area that stores the pixel data of an image before it is displayed on the screen.

Q4 How many Bits are used in Bitmap?

Ans It depends on the color depth of bitmap, such as 1-bit, 8-bit, 16-bit, 24-bit or 32-bit.
More bits allow more colors to be represented

Q5 What is Pixmap?

A pixmap (pixel map) is a representation of an image where each pixel is stored with its color information.
It is similar to a bitmap but can support different pixel / color formats.

Date

Expt. No. 2. Page No. 3

Aim:- To implement features like changing background color, fore ground color, resizing window, repositioning of window.

Source Code:-

```
main() {
    int gd = DETECT, gm;
    initgraph (&gd, &gm, "C:\\TC\\BGI");
    midx = getmaxx()/2;
    midy = getmaxy()/2;
    setbkcolor (GREEN);

    setviewport (midx - 50, midy - 50, midx + 50,
                 midy + 50, 1);
    circle (100,100,50);    /* drawn in white */
    setcolor (RED);
    circle (200,200,50);    /* drawn in red */
    getch();
    closegraph();
    return 0;
}
```

Viva Question

Q1 What is clipping?

Ans Clipping is the process of removing the parts of an object that lie outside the specified viewing area. It helps display only the visible portion of the object.

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Q2 Difference between Raster Scan & Random Scan.

Ans

Raster Scan	Random Scan
• Scans the screen line by line from top to bottom	Draws only the required lines of an image.
• Suitable for realistic images & filled areas	Suitable mainly for line drawings.
• Uses a frame buffer	Uses a display list.

Q3 How many bits are required for a 512 * 512 raster with each pixel being represented by 3 bits.

Ans Number of pixels = 512 x 512 = 262144 pixels

Bits required : 262144 x 3 = 786,432 bits
= 98304 bytes = 96 KB

So required bits are 96 KB.

Q4 Define Point clipping?

Ans Point clipping determines whether a point lies inside or outside the clipping window. If the point is inside, it is displayed; otherwise, it is discarded.

Q5 In a raster system with resolution 2560 x 2048. How many pixels could be accessed per second by a display controller that refresh the screen

at a rate of 60 frames per seconds. Also calculate access time per pixel in the system

Ans

Total Pixels = 2560 × 2048 = 5242880 Pixels.

Pixels accessed per second

$5242880 \times 60 = 314572800$ pixel/sec

Access time per pixel:

$$\frac{1}{314572800} \approx 3.18 \times 10^{-9} \text{ second.}$$

$\approx$ 3.18 nS per

So

Pixel/second = 314572800 Pixels/sec

Access time/pixel = 3.18 nanosecond.

Aim :- Write a program to implement Bresenham line Algorithm.

Source Code

```
#include <stdio.h>
#include <conio.h>
#include <graphics.h>
void main()
{
  int gd = DETECT, gm;
  int dx, dy, p, end;
  float x1, x2, y1, y2, x, y;
  initgraph(&gd, &gm, "c:\tc\bgi");

  printf("Enter value of X1 : ");
  scanf("%f", &x1);
  printf("Enter value of Y1 : ");
  scanf("%f", &y1);
  printf("Enter value of X2 : ");
  scanf("%f", &x2);
  printf("Enter value of Y2 : ");
  scanf("%f", &y2);

  dx = abs(x1 - x2);
  dy = abs(y1 - y2);
  p = 2 * dy - dx;
```

```
if(x1>x2)
 {
 x = x2;
 y = y2;
 end = x1;
 }
else {
 x = x1;
 y = y1;
end = x2;
 }
putpixel (x, y, 10);
while (x < end).
 {
   x = x+1;
   if (p<0)
    p = p + 2*dy;
    }
   else
     { y = y+1;
       p = p + 2*(dy-dx);
     }
   putpixel (x, y, 10);
 }
getch();
closegraph();
}
```

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Viva Question

Q1 What are the advantage of Bresenham's Algorithm?

Ans Brensenham's algorithm is fast & efficient because it uses only integer arithmetic, avoiding floating-point calculations. It produces accurate lines & requires less computation time & memory.

Q2 What do you call the path the electron beam takes when returning to left side of CRT screen?

Ans The path taken by the ~~select~~ electron beam when it returns to the left side of the CRT screen is called horizontal retrace. During this process, the beam returns to the ~~string~~ starting position for the next line.

Q3 What is the refresh rate of Flicker?

Ans The refresh rate is the number of times a screen images is redrawn per second, measured in Hertz (Hz). Flicker can be reduced by using a higher refresh rate, typically 60Hz or above.

Q4 Write short note on active & passive transformation

Ans Active transformation changes an object's position or orientation relative to a fixed coordinate system.

Passive transformation changes the coordinate system while the object remains physically unchanged.

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Date

Expt. No. _____ Page No. 8.

Q5 What do you call the path the electron beam takes at the end of each refreshed cycle?

Ans The path taken by the electron beam at the end of each refresh cycle is called vertical retrace. During this process, the beam returns from the bottom of screen to the top to begin the next refresh cycle.

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Date

Expt. No. 4. Page No. 9

Aim :- Write a program to implement DDA line algorithm.

Source Code

```
#include <stdio.h>
#include <conio.h>
#include <graphics.h>
#include <ctype.h>
#include <math.h>
#include <stdlib.h>

void draw(int x1, int y1, int x2, int y2);
void main()
{
int x1, x2, y1, y2;
int gdriver = DETECT, gmode, errorr;
initgraph(&gdriver, &gmode, "c:\\tc\\bgi");

printf("\n Enter the x & y value for starting
        point:\n");
scanf("%d%d", &x1, &y1);
printf("\n Enter the x & y value for ending
        point:\n");
scanf("%d%d", &x2, &y2);
printf("\n The line is shown below:\n");
draw(x1, y1, x2, y2);
getch();
}
```

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```
void draw (int x1, int y1, int x2, int y2)
{
float x, y, xinc, yinc, dx, dy;
 int k;
int step;
dx = x2 - x1;
dy = y2 - y1;
if (abs(dx) > abs(dy))
 step = abs(dx);
else
 step = abs(dy);
xinc = dx/step;
yinc = dy/step;

x = x1;
 y = y1;
putpixel (x1, y1, 1);
for (k=1; k <= step; k++)
 {
 x = x + xinc;
 y = y + yinc;
 putpixel(x, y, 2);
 }
}
```

Viva Questions

Q1. What is DDA?

Ans DDA (digital differential Analyzer) is a line-drawing ~~drawing~~ algorithm used in computer graphics to draw a straight line between two points using incremental values of x & y.

Q2. What is run length encoding?

Ans Run length Encoding (RLE) is a data compression technique that replace consecutive repeated values with the value & its count. It reduces storage space, espacially when the data contains many repeated characters or pixels.

Q3 What is the need of homogeneous coordinates?

Ans Homogeneous Coordinates represents 2D points using three values $(x, y, 1)$. They simplify geomatric transformations by allowing translation, rotation & scaling to be represented using matrix multiplications.

Q4 What are the disadvantages of DDA Algorithm?

Ans DDA may produce rounding errors because it uses floating-point calculations. It is also generally slower & less accurate than Bresenham's line-drawing algorithm, which uses integer arithmetic.

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Q5. What is Translation?

Ans Translation is a geometric transformation that moves an Object from one position to another by adding fixed values to its coordinates. It changes the position of an object without changing its shape, size or orientation.

Aim:- Write a program to implement midpoint circle algorithm

Source Code

```
#include <stdio.h>
#include <conio.h>
#include <graphics.h>
void main()
  {
  int gd = DETECT, gm;
  int x, y, r;
  void Drawcircle (int, int, int);
  printf("Enter the mid points & Radius:");
  scanf("%d %d %d", &x, &y, &r);
  initgraph(&gd, &gm, "");
  Drawcircle(x, y, r);
  getch();
  closegraph();
}
void Drawcircle(int x, int y, int r)
  {
  int x=0, y=r, p=1-r;
void clipot(int, int, int, int);
clipot(x, y, x, y);
while(x<y)
  { x++;
    if(p<0)
      p+= 2*x+1;
    else { y--;
           p+= 2*(x-y)+1; } }
```

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```
        clipbot(x1, y, x, y);
        }
    }
void clipbot(int xctr, int yctr, int x, int y)
    {
putpixel(xctr+x, yctr+y, 1);
putpixel(xctr-x, yctr+y, 1);
putpixel(xctr+x, yctr-y, 1);
putpixel(xctr-x, yctr-y, 1);
putpixel(xctr+y, yctr+x, 1);
putpixel(xctr-y, yctr+x, 1);
putpixel(xctr+y, yctr-x, 1);
putpixel(xctr-y, yctr-x, 1);
    getch();
}
```

Viva Question

Q1 Define Circle.

Ans A circle is a set of points in a 2D plane that are at a fixed distance, called the radius, from a fixed point called the center. In Computer graphics, circle can be drawn using algorithms like Bresenham's circle Algorithm.

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~~Viva Questions~~

Q2 How many memory bits are required for a 24-bit plane 1024 x 1024 element raster?

Ans Memory required = 1024 x 1024 x 24 bit

Therefore, the total memory required is 25,165,824 bits (3MB).

Q3 What is Rotation?

Ans Rotation is a geometric transformation that turns an object around a fixed point through a specified angle. It changes the orientation of the object without changing its shape / size.

Q4 Distinguish between uniform scaling & differential scaling?

Ans Uniform scaling changes the size of an object equally in all directions using the same scaling factor. Differential scaling factors for different axes changing the object's proportions.

Q5 What is Back Face Removal?

Ans Back face removal is a technique in computer graphics used to eliminate the faces of a 3D object that are not visible to the viewer. It improves rendering efficiency by avoiding the drawing of unnecessary surfaces.

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